

Note on Over-simplified Cross-Situational Learning Models

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Abstract

1 Blythe, Smith, & Smith (2010)

1.1 On the fast-mapping learner

Suppose that the fast-mapping learner learns a word independently drawn from uniform distribution of W words given in each episode. As every episode has one word by the probability $1/W$, this is identified with so called Coupon Collector's Problem. In this problem, the expectation of the time to sample all the words T is

$$E[T] = \sum_{i=1}^W E[t_i].$$

where t_i is the time to sample a i^{th} new word given $(i - 1)$ words being learned. Thus,

$$T = W \sum_{i=1}^W 1/i \approx W \log W.$$

1.2 Extension of the fast-mapping model

This estimate by Blythe et al. is a quite “optimistic”, as the target word is sampled uniformly. Now let us consider a more realistic case – the word

is drawn from a Zipf's distribution (or power distribution). Suppose that the i^{th} word is drawn by the probability $p_i \geq 0$, and there are n words and $\sum_{i=1}^n p_i = 1$. Write $p = (p_1, \dots, p_n)$. According to Hidaka (2014), the probability to have the number of unique words (types) k for m sampled words (tokens) is, as $m \rightarrow \infty$,

$$P(k \mid m, p) \approx Q_k(p_m),$$

where $Q_k(p_m)$ is the Poisson-binomial distribution with the parameter $p_m := (p_{1,m}, p_{2,m}, \dots, p_{n,m})$ and $p_{i,m} := 1 - (1 - p_i)^m$.

Suppose, for a small $\epsilon > 0$, the probability to complete sample n types satisfies

$$\prod_{i=1}^n p_{i,m} > 1 - \epsilon.$$

Using this inequality with $\epsilon = 0.01$, which Blythe et al assumed in their estimate, we have the results for example:

1. (Uniform sampling) For the set of $n = 10,000$ words drawn by the uniform distribution $p_i = 1/n$ and $\epsilon = 0.01$, the minimal m satisfies the inequality above is 46,050,138,098.
2. (Zipf's law) For the set of $n = 10,000$ words drawn by the Zipf's distribution $p_i \propto 1/i$ and $\epsilon = 0.01$, the minimal m satisfies the inequality above is 1,099,950.

The first case is considered in Blythe et al. (2010), and they argued that this is a reasonable estimate with respect to empirical findings in developmental literature. The second case, in which I additionally relax their uniform sampling assumption, offers a "less reasonable" estimate, which is over 20 times more number of samples to finish learning than the uniformly distributed case. This estimate may break down their theory: As the fast-mapping estimate is the minimal baseline required to learn words, their assumption that independent learning is shown quite unreasonable to begin with.

2 Appendix

For a set of N words drawn from the distribution $p = (p_1, \dots, p_N)$, the proportion of children finished learning all the words is $(1 - \epsilon)$ for $0 < \epsilon < 1$

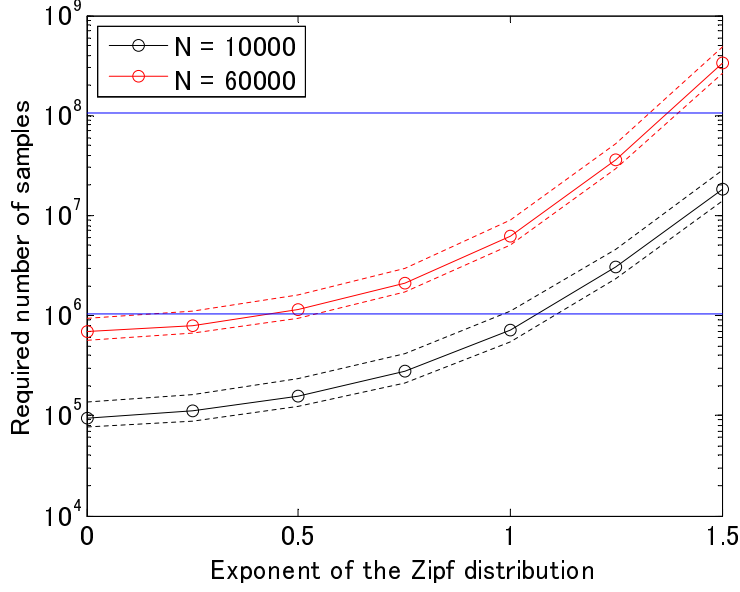


Figure 1: For $\epsilon = 0.01, 0.5, 0.99$ (broken and solid lines), $N = 10000, 60000$ and the exponent $a = 0, 0.25, \dots, 1.5$, the required number of samples M for a generalized Zipf distribution $p_k = k^{-a} / \sum_{k=1}^N k^{-a}$ is numerically calculated by the root of Equation (1).

requires the sample size M , which is the root of

$$\prod_{i=1}^N (1 - (1 - p_i)^M) = 1 - \epsilon. \quad (1)$$

Write

$$f_{N,\epsilon}(x) := \frac{\log(1 - (1 - \epsilon)^{1/N})}{\log(1 - x)}.$$

In a special case with a set of N words distributed uniformly, the proportion of children finished learning all the words is $(1 - \epsilon)$ requires the sample size

$$M = f_{N,\epsilon}(1/N).$$

By the intermediate value theorem, the root of (1) is expressed with some

53 constant $\min p \leq c \leq \max p$

$$M = f_{N,\epsilon}(c).$$

54 Equivalently, we have inequality

$$f_{N,\epsilon}(\max p) \leq M \leq f_{N,\epsilon}(\min p).$$

55 Write $M_+ = f_{N,\epsilon}(\min p)$, which gives a reasonable estimate of the upper
 56 bound of the root M of (1). For the Zipf distribution with N words and
 57 exponent a , $\min p = N^{-a} H_{N,a}^{-1}$ where $H_{N,a} := \sum_{k=1}^N k^{-a}$ is the generalized
 58 harmonic number.

$$\frac{\partial f_{N,\epsilon}(x)}{\partial x} = -\frac{\log(1 - (1 - \epsilon)^{1/N})}{(1 - x) \log(1 - x)^2} = -\frac{f_{N,\epsilon}(x)}{(1 - x) \log(1 - x)}.$$

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$$\min p = \frac{N^{-a}}{\sum_{k=1}^N k^{-a}}$$

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$$\frac{\partial}{\partial a} \min p = \frac{-N^{-a} \log N \left(\sum_{k=1}^N k^{-a} \right) - N^{-a} \left(-\sum_{k=1}^N k^{-a} \log k \right)}{\left(\sum_{k=1}^N k^{-a} \right)^2}$$

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$$\frac{\partial}{\partial a} \min p = \min p \frac{\left(\sum_{k=1}^N k^{-a} \log \frac{k}{N} \right)}{\sum_{k=1}^N k^{-a}} = \min p \left(-\frac{\partial H_{N,a}}{\partial a} / H_{N,a} - \log N \right)$$

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$$\frac{\partial \log M_+}{\partial a} = \frac{p_{\min} \left(\frac{\partial H_{N,a}}{\partial a} / H_{N,a} + \log N \right)}{(1 - p_{\min}) \log(1 - p_{\min})}$$

63 and

$$\frac{\partial^2 \log M_+}{\partial a^2} \geq 0.$$

64 This implies the M_+ is a super-exponential monotone function.

65 2.1 Cross-situational learning

66 The idea used in the analysis of fast learning can be generalized to analyze
 67 the cross situation learning.

68 Suppose, for a particular target word in a set of N words, i^{th} other
 69 word appears by the probability q_i in each learning episode. Write $q =$

70 $(q_1, \dots, q_{N-1})$. The probability to none of the $N - 1$ other words accompanies
 71 with the target word for every time of k learning episodes is

$$P_{\text{XSL}}(K, q) = \prod_{i=1}^{N-1} (1 - q_i^K).$$

72 Find this equation is essentially identical to the left hand side of (1) by
 73 identifying $K = M$ and $q_i = 1 - p_i$. Thus, the upper bound of the K is given
 74 by

$$K_+ = f_{N-1, \epsilon}(1 - \max q).$$

75 Putting together, the time for cross situational learning is bounded by Putting
 76 together, the time for cross situational learning is bounded by

$$T_+ = \max_i \frac{f_{N-1, \epsilon}(1 - \max q_i)}{-\log(1 - p_i)} \approx \frac{\log(1 - (1 - \epsilon)^{1/N})}{\min(\log(1 - p_i) \log(\max q_i))}.$$

77 This analysis implies the following.

- 78 1. The uniform sampling of (1) target word and (2) incidental words gives
 79 shortest learning time.
- 80 2. The sampling of (1) rarely drawn target word accompanying with (2)
 81 highly likely incidental words gives longest learning time.

82 It is obvious that the uniform sampling gives the largest $\min p$ and smallest
 83 $\max q$. On the other hand, a class of Zipf distributions give relatively severe
 84 conditions for both fast mapping and cross situational learning, as it gives
 85 extreme values – smaller $\min p$ and larger $\max q$. Specifically, under the
 86 Zipf distribution with exponent a , the learning time for the worst case 2. is
 87 bounded by the super-exponential function of the a .

88 In summary, this analysis revealed that efficient cross situational learning
 89 needs to have every target word and its incidental distribution is near uniform
 90 distributions in both p and q . This condition required for the efficient word
 91 learning is hard to believe in with respect to empirical data (Figure 2)!

$$P_{\text{XSL}}(m_i, q_i) = \prod_{j \neq i} (1 - q_{i,j}^{m_i}).$$

92

$$P_{\text{XSL}}(M, p, q) = \sum_{\sum_{i=1}^N m_i = M} P_{\text{XSL}}(m_i, q_i) \frac{M!}{\prod_{i=1}^N m_i} \prod_{i=1}^N p_i^{m_i}$$

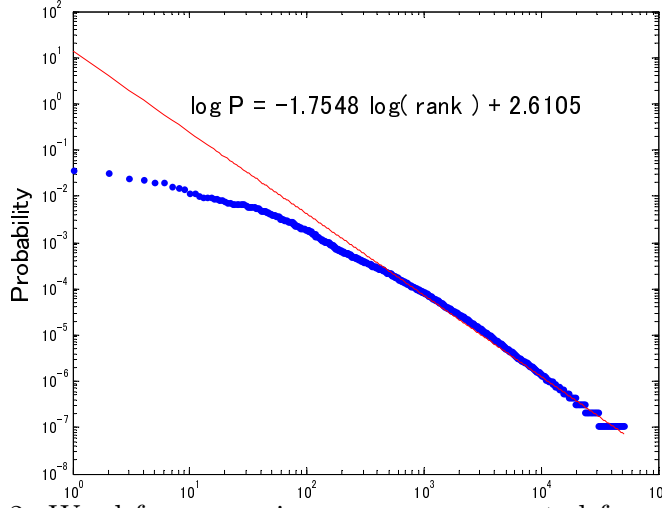


Figure 2: Word frequency in Rank order as aggregated from the CHILDES transcripts.

3 Relative frequency coding

Obviously, the eliminative learning, which relies on the assumption that the target word is always spoken in the present of the corresponding object, is quite limited and not realistic with respect to child learning. Such an eliminative learner failed to learn even with an object without being associated to its word.

Let us consider a learner who learns word-object pair on basis of relative frequency of words given an object. Suppose the word 1 is the target word for the object 1, and let p_i be the probability to have word i associated to the object 1. Suppose the probability to have the target word is $p_1 > p_i$ for $1 < i \leq n$, and the learner draws m object samples, in which n words are drawn independently from the distribution $(p_1, \dots, p_n)$. For m object samples, let $(m_1, \dots, m_n)$ be the vector of the cumulative counts of words. The relative frequency learner estimate, the target word is i for the object 1, if $m_i > m_j$ for $j \neq i$. This relative frequency learning is more robust to noisy object without its target word, and the learning can be completed in a long run as long as the target word is more likely than the other words.

110 Then the question is how many samples this relative frequency learner
 111 needs to complete learning. For word i drawn by the probability p_i for each
 112 sample, the expectation of the cumulative count is $E[m_i] = mp_i$, and its
 113 variance is $V[m_i] = mp_i(1 - p_i)$. According to the Chebyshev's inequality,
 114 for a constant $a > 0$, we have

$$P(m_i \in I_{i,a}) > 1 - \frac{V[m_i]}{a^2}.$$

115 Let us write the interval by $I_{i,a} := \{x : |x - E[m_i]| < a\}$. Thus it im-
 116 plies that the cumulative count m_i is on the interval $I_{i,a}$ by an arbitrar-
 117 ily high probability for sufficiently large m $\left(\lim_{m \rightarrow \infty} \frac{V[m_i]}{a^2}\right)$ by setting $a :=$
 118 $\sqrt{a_i(m)mp_i(1 - p_i)}$, where $a_i(m)$ is a function satisfying $\lim_{m \rightarrow \infty} \frac{a_i(m)}{m} = 0$
 119 and $\lim_{m \rightarrow \infty} \frac{1}{a_i(m)} = 0$. Then, for sufficiently large m , we have the proba-
 120 bility for the relative frequency learner to estimate the correct target word
 121 is

$$P(m_1 > \max(m_2, \dots, m_n)) \geq 1 - \left(1 - \prod_{i=2}^n \frac{mp_i(1 - p_i)}{a_i^2}\right) \frac{mp_1(1 - p_1)}{a_1^2}. \quad (2)$$

122 For the threshold ϵ and the probability $p_{i,j}$ of the word j given the object i ,
 123 the required number of sample m to learn the word 1 satisfies the inequality

$$\prod_{i=1}^n \left\{ 1 - \left(1 - \prod_{j \neq i} \frac{mp_{i,j}(1 - p_{i,j})}{a_{i,j}^2} \right) \frac{mp_{i,i}(1 - p_{i,i})}{a_{i,i}^2} \right\} = 1 - \epsilon. \quad (3)$$

124 Find the similarity of this to the equation (1). The term in (3) corresponding
 125 to the term $(1 - p_i)^M$ decaying exponentially as a function of M in Equation
 126 (1) is as slow as a power function of m .

127 Acknowledgment

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